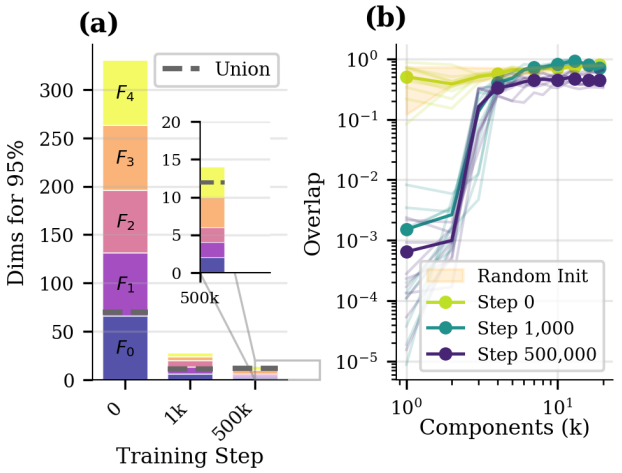

Figure 3. **Transformers represent factors in orthogonal multi-dimensional subspaces.** (a) The number of dimensions needed to reach a CEV of 95% in the residual stream at the output layer of the last transformer block when driven by data from the vary-one dataset. At initialization, the activations associated with each factor use approximately the same number of dimensions as the union suggesting the subspaces are effectively fully overlapping. Over training the sum of the factor's dimensions approaches the dimension of the union, which suggests orthogonality. (b) Indeed, the first two principal components (PCs) associated with each factor evolve over training to become orthogonal to those associated with other factors. Thin lines show the normalized subspace overlap between the PCs belonging to different factors, while the heavy lines are the average over all pairs at each training step. The highlight shows a 90% confidence interval for initial factor overlap, generated from an ensemble of randomly initialized models.

our earlier experiments—but now each of the subprocesses depends on all of the factors above it in the chain.

We find that neural networks trained via next-token prediction on the resulting sequences automatically learn to factor their representation into parts, where uncertainty over the latent states of each part is encoded geometrically. See Appendix. D.2 for additional details.

4.3. Factored representations as an inductive bias

The previous experiments demonstrate that transformers learn factored representations when factoring is valid. But this leaves open a crucial question: is factoring merely a sufficient solution that the model stumbles upon, or is it a preferred solution that the architecture actively seeks?

To distinguish these possibilities, we need to examine cases where factoring is *not* valid—where the factored representation *is* lossy, and the model must choose between representing the world as made of parts and predictive fidelity.

Controlled violations of conditional independence. We take our five independent factors and pass the token through a noisy channel. With probability $1 - \varepsilon$, the token is transmitted faithfully; with probability ε, it is replaced by a uniformly random token. For example, at $\varepsilon = 0.5$, there is a 50% chance that the observed token reflects the outputs of the five factors, and a 50% chance it is pure noise. This simple corruption breaks conditional independence. When the channel transmits faithfully, observing the token updates information about each factor independently. But when the channel corrupts, the observation provides no information, and critically, the observer doesn't know which case occurred. The result is that beliefs about different factors become correlated in ways that cannot be factored. Predictive vectors drift off the product-state manifold, and projecting onto the factored representation discards relevant information about these cross-factor correlations.

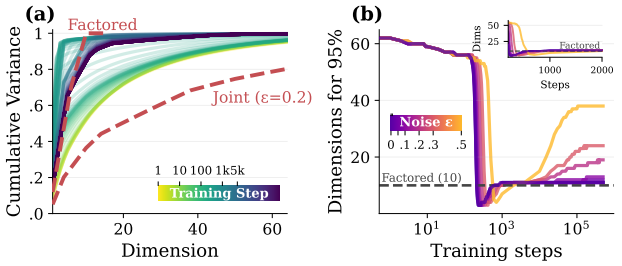

Figure 4. **Transformers have an inductive bias towards factored representations.** Even when the generative process doesn't admit a lossless factorization (as is the case for nonzero values of ε), the transformer quickly learns one anyways. For small ε, this factored representation persists throughout training. As ε grows, transformers find the factored form early before expanding their dimensionality for greater predictive fidelity.

This creates a measurable tradeoff. The factored representation offers dimensional efficiency at the cost of predictive fidelity. The joint representation offers fidelity at the cost of dimensionality. A model that treats factoring as merely sufficient would use whatever representation minimizes loss. A model with an inductive bias toward factoring would prefer the factored solution even when it hurts.

Explained Variance over Training. Figure 4(a) shows the CEV curves over training for $\varepsilon = 0.2$, when the noisy channel has a 20% chance of randomizing the hidden tokens (See Figure 12 for other ε analysis). Over training, activations compress and converge close to the dimensionality of the factored representations we find in Figures 2 and 3. Notably, this happens despite the fact that the factored representation is lossy.

Factored reps are preferred even when non-optimal. Figure 4(b) shows the number of principal components required to capture 95% of activation variance over training, for training runs at several values of ε. Remarkably, regardless of ε, transformers first collapse to the 10-dimensional factored representation within hundreds of gradient steps.

Only later, under sustained pressure from the loss gradient, do they gradually expand into higher dimensions.

This is not what we would expect if the model were simply finding the lowest-loss solution. At high ε, the factored representation leaves substantial predictive information on the table, information the model demonstrably learns to recover later in training. Yet the model dwells in the factored regime first, departing only reluctantly.

The factored solution acts as a representational attractor. Models find it quickly, dwell there, and depart only under sustained gradient pressure. This suggests that factorization is not merely a capacity-saving trick for tractable problems, but a genuine inductive bias of the architecture.

4.4. RNNs also learn to factor

We ran the same experiments and analysis on vanilla RNNs and LSTMs to check whether our results are particular to transformers. We find that they partially generalize: in all cases, we find CEV curves that are more consistent with a factored representation than a joint one, though with the precise shape varying based on d_{model}. All tested models contain linear maps from their activations to all 5 factors. For RNNs, at lower d_{model}, we find that the CEV curves closely match our Transformer results, with 10 components explaining 95% of the cumulative variance. As we scale up d_{model}, the CEV curves smooth out even when the model does not have capacity for the joint representation. For LSTMs, we observe this phenomenon at smaller d_{model} sizes. See Appendix. D.4 for more details.

5. Related work

There have been many efforts to *engineer* factored representations (Degris & Sigaud, 2013; Sallans, 1999). For example, RL has been used to learn factored Markov decision processes (Guestrin et al., 2003; Strehl et al., 2007), and much work has gone into learning the structure of dynamic Bayesian networks (Kaddour et al., 2022). In contrast, our results investigate how factorization *naturally* arises through pretraining. Specifically, we observe that neural networks discover relevant factored belief representations without any explicit network, graph, or reinforcement learning.

New architectures have been proposed to force factored representations (Dinh et al., 2014; Nanbo et al., 2025). In our experiments, we find that neural networks do this during next-token pretraining, with no new architecture needed.

In the field of interpretability, our work is consistent with recent work that also finds multi-dimensional geometric relationships among activations in LLMs (Engels et al., 2025; Gurnee et al., 2025; Kantamneni & Tegmark, 2025). A novel contribution of our work is its constructive nature: we predict what these higher dimensional features *should* be. These multi-dimensional belief geometries are dense in their active subspaces, which may help explain why some sparse-autoencoder (SAE) latents are dense (Sun et al., 2025). This invalidates the explicit assumption motivating SAEs (Elhage et al., 2022), that all concepts would be orthogonal if given sufficiently many dimensions. We find that nonorthogonality of activations within the factor's subspace purposefully encodes similar predictions. This may enable some ability to anticipate and interpret potentially universal context embeddings (Huh et al., 2024; Sucholutsky et al., 2023).

When representations factor into orthogonal subspaces, it allows an understanding of which behaviors can be tuned independently, enabling rather surgical interventions (Feucht et al., 2025). It also hints at how fine tuning on one aspect of behavior in one context can enhance or suppress that behavior across others (Betley et al., 2025).

The emergence of belief geometry from pretraining was first found by Shai et al. (2024), and its mechanisms are just beginning to be understood (Piotrowski et al., 2025; Riechers et al., 2025). The present contribution shows that factoring beliefs into conditionally independent parts allows further dimensionality reduction in representations of Bayesian beliefs over generative world models.

Boyd et al. (2025) describe the information theory of factorizing processes into modular subcomponents, showing how the composition of stochastic automata (Hopcroft & Ullman, 1979; Mohri, 1997; Mohri et al., 2002) can be inverted. Our work suggests neural networks automatically learn a representational version of this causal graph discovery (Nogueira et al., 2022), similar to transformers' ability to learn Krohn–Rhodes and other decompositions to efficiently simulate deterministic automata (Liu et al., 2023).

Our work brings reason for representational optimism (Kumar et al., 2025), and may explain disentangled representations in models' subnetworks (Zhang et al., 2021).

6. Conclusion

In this work, we discovered that, as a natural consequence of next-token pretraining, neural networks automatically learn to factor the world into parts. To make rigorous falsifiable claims, we trained relatively small transformers on various sources of correlated sequences designed to test when and how these models factor their activations.

We found that we could precisely anticipate the learned representations—how many dimensions of the residual stream would be used, how many multi-dimensional subspaces would be updated in parallel, and even the precise geometric pattern of non-orthogonal activations induced by different contexts within each subspace. We found an induc-

tive bias for transformers to factor their internal activations early in training, even when this sacrifices predictive fidelity. We ran analogous experiments on RNN architectures, and found that they too learn these factored representations, hinting at the universality of these results.

Given the theoretical underpinnings of our empirical results, it is plausible that the the factored world hypothesis extrapolates to larger models as well. [4] This sets the stage for many new opportunities in interpretability: rather than needing to understand the entire neural network all at once, this gives hope that we can discover some low-dimensional interpretable subspaces in the residual stream. New interpretability tools will be needed to discover these factored-representation subspaces in an unsupervised fashion.

We could then trace information flow within and amongst these subspaces, each a separable part of the cognitive whole. With this computational universe unentangled into its basic constituents, we may then hope to build a fundamental psychology, respecting the structure of concepts that models natively represent.

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References

Betley, J., Tan, D., Warncke, N., Sztyber-Betley, A., Bao, X., Soto, M., Labenz, N., and Evans, O. Emergent misalignment: Narrow finetuning can produce broadly misaligned llms. *arXiv preprint arXiv:2502.17424*, 2025.

Boyd, A., Nowak, F., Hyland, D., Baltieri, M., and Rosas, F. E. From monoliths to modules: Decomposing transducers for efficient world modelling. *arXiv preprint arXiv:2512.02193*, 2025.

Degris, T. and Sigaud, O. *Factored Markov Decision Processes*, chapter 4, pp. 99–126. John Wiley & Sons, Ltd, 2013. ISBN 9781118557426. doi: https://doi.org/10.1002/9781118557426.ch4. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118557426.ch4.

Dinh, L., Krueger, D., and Bengio, Y. Nice: Non-linear independent components estimation. *arXiv preprint arXiv:1410.8516*, 2014.

[4] Indeed, for frontier models trained on such massive datasets, there will be extreme pressure to factor the learned world model into parts since this yields the exponential dimensional advantage that the model would need to capture much of anything in the latent space of its finite model dimension.

Elhage, N., Hume, T., Olsson, C., Schiefer, N., Henighan, T., Kravec, S., Hatfield-Dodds, Z., Lasenby, R., Drain, D., Chen, C., et al. Toy models of superposition. *arXiv preprint arXiv:2209.10652*, 2022.

Engels, J., Michaud, E. J., Liao, I., Gurnee, W., and Tegmark, M. Not all language model features are one-dimensionally linear. *ICLR 2025, arXiv:2405.14860*, 2025.

Fanizza, M., Lumbreras, J., and Winter, A. Quantum theory in finite dimension cannot explain every general process with finite memory. *Communications in Mathematical Physics*, 405(2):50, 2024.

Feucht, S., Wallace, B., and Bau, D. Vector arithmetic in concept and token subspaces. *arXiv preprint arXiv:2511.18162*, 2025.

Guestrin, C., Koller, D., Parr, R., and Venkataraman, S. Efficient solution algorithms for factored mdps. *Journal of Artificial Intelligence Research*, 19:399–468, 2003.

Gurnee, W., Ameisen, E., Kauvar, I., Tarng, J., Pearce, A., Olah, C., and Batson, J. When models manipulate manifolds: The geometry of a counting task. *Transformer Circuits Thread*, 2025. URL https://transformer-circuits.pub/2025/linebreaks/index.html.

Hopcroft, J. E. and Ullman, J. D. *Introduction to Automata Theory, Languages, and Computation*. Addison-Wesley, Reading, 1979.

Huh, M., Cheung, B., Wang, T., and Isola, P. The platonic representation hypothesis. *arXiv preprint arXiv:2405.07987*, 2024.

Kaddour, J., Lynch, A., Liu, Q., Kusner, M. J., and Silva, R. Causal machine learning: A survey and open problems. *arXiv preprint arXiv:2206.15475*, 2022.

Kantamneni, S. and Tegmark, M. Language models use trigonometry to do addition. *arXiv preprint arXiv:2502.00873*, 2025.

Kingma, D. P. and Ba, J. Adam: A method for stochastic optimization. *CoRR*, abs/1412.6980, 2014. URL https://api.semanticscholar.org/CorpusID:6628106.

Kumar, A., Clune, J., Lehman, J., and Stanley, K. O. Questioning representational optimism in deep learning: The fractured entangled representation hypothesis. *arXiv preprint arXiv:2505.11581*, 2025.

Liu, B., Ash, J. T., Goel, S., Krishnamurthy, A., and Zhang, C. Transformers learn shortcuts to automata. *ICLR*, 2023.